

# LAB OBSERVATION

## EXPERIMENT-12

Implement an Earley parser for context-free grammars using a simple python program.

### CODE:

```
[2] ✓ 9%  
1 import nltk  
2 from nltk import CFG  
3 from nltk.parse import EarleyChartParser  
4  
5 grammar = CFG.fromstring("""  
6 S -> NP VP  
7 NP -> 'John'  
8 VP -> V NP  
9 V -> 'sees'  
10 """)  
11  
12 parser = EarleyChartParser(grammar)  
13  
14 sentence = input("Enter sentence: ").split()  
15  
16 for tree in parser.parse(sentence):  
17     print(tree)
```

variables Terminal 2:00 PM Python 3  
36°C Mostly sunny 23-07-2026

### OUTPUT:

```
... Enter sentence: John sees John  
(S (NP John) (VP (V sees) (NP John)))
```

Variables Terminal 2:00 PM Python 3  
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